

天津大学

《智能搜索与问答》实验报告



Lab6 信息检索系统

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一、 数据准备与处理

1.1 数据集

数据集使用的是简单英语维基百科（Simple English Wikipedia），与完整的英文维基百科相比，它只有大约 170k 篇文章。数据集下载地址为 <http://sbert.net/datasets/simplewiki-2020-11-01.jsonl.gz>

1.2 数据集预处理

对于数据集，我们将这些文章分成段落，并使用双编码器对它们进行编码。

```
wikipedia_filepath = 'data/simplewiki-2020-11-01.jsonl.gz'

if not os.path.exists(wikipedia_filepath):
    util.http_get('http://sbert.net/datasets/simplewiki-2020-11-01.jsonl.gz', wikipedia_filepath)

passages = []
with gzip.open(wikipedia_filepath, 'rt', encoding='utf8') as fIn:
    for line in fIn:
        data = json.loads(line.strip())

        #Add all paragraphs
        #passages.extend(data['paragraphs'])

        #Only add the first paragraph
        passages.append(data['paragraphs'][0])

print("Passages:", len(passages))

corpus_embeddings_path = 'data/corpus_embeddings.pt'
if os.path.exists(corpus_embeddings_path):
    corpus_embeddings = torch.load(corpus_embeddings_path)
else:
    logging.warning("Prebuilt corpus_embeddings not found. Please run the next block to compute your own corpus embedding")
```

图 1. 数据集预处理 1

我们将所有段落编码到我们的向量空间中。这种方法只适合那些想从头开始计算自己的语料库嵌入的人。

```
# NOTE: only for those who want to compute your own corpus embeddings from scratch
# We encode all passages into our vector space.
corpus_embeddings = bi_encoder.encode(passages, convert_to_tensor=True, show_progress_bar=True)
```

图 2. 数据集预处理 2

二、 模型与算法介绍

2.1 任务介绍

本实验是 Retrieve & Re-Rank Demo over Simple Wikipedia。该实验实现了检索和重新排名设置，并允许在简单维基百科[Simple Wikipedia](https://simple.wikipedia.org/wiki/Main_Page)上进行搜索。

我们可以输入查询或问题，然后脚本使用语义搜索在简单维基百科上找到相关段落。

2.1 模型结构

对于语义搜索，我们使用 'multi-qa-distilbert-cos-v1' 模型并检索 32 个可能回答输入查询的段落。

SentenceTransformers 是一个可以用于句子、文本和图像嵌入的 Python 库。可以为 100 多种语言计算文本的嵌入并且可以轻松地将它们用于语义文本相似性、语义搜索和同义词挖掘等常见任务。该框架基于 PyTorch 和 Transformers，并提供了大量针对各种任务的预训练模型。还可以很容易根据自己的模型进行微调。

multi-qa-distilbert-cos-v1 模型是一个 SentenceTransformers，他将它将句子和段落映射到一个 768 维的密集向量空间，并被设计用于语义搜索。它已经在来自不同来源的 215M（问题、答案）对上进行了训练。

```
from sentence_transformers import SentenceTransformer, util

query = "How many people live in London?"
docs = ["Around 9 Million people live in London", "London is known for its financial district"]

#Load the model
model = SentenceTransformer('sentence-transformers/multi-qa-distilbert-cos-v1')

#Encode query and documents
query_emb = model.encode(query)
doc_emb = model.encode(docs)

#Compute dot score between query and all document embeddings
scores = util.dot_score(query_emb, doc_emb)[0].cpu().tolist()

#Combine docs & scores
doc_score_pairs = list(zip(docs, scores))

#Sort by decreasing score
doc_score_pairs = sorted(doc_score_pairs, key=lambda x: x[1], reverse=True)

#Output passages & scores
for doc, score in doc_score_pairs:
    print(score, doc)
```

图 3. SentenceTransformer 代码示例

接下来，我们使用更强大的 CrossEncoder ('qnli-distilroberta-base' 模型)，它对查询和所有检索到的段落的相关性进行评分。交叉编码器进一步提升了性能，尤其是当搜索未训练双编码器的语料库时。

cross-encoder 是在模型输入时进行了拼接输入，这样两个问句进行更深层的交互，相似句任务直接在模型中完成，而不仅仅是问句的语义表示模型。所以 **cross-encoder** 的相似计算效果也要明显优于双塔结构的 **bi-encoder**，但是 **cross-encoder** 不能得到问句的向量表示，大量问句间的相似计算需要实时进行模型推理计算而消耗的时间更多。**cross-encoder** 适合用于少量候选问句场景，因此在选择数据集时选择的是 Simple Wikipedia，数据集规模不是特别大。

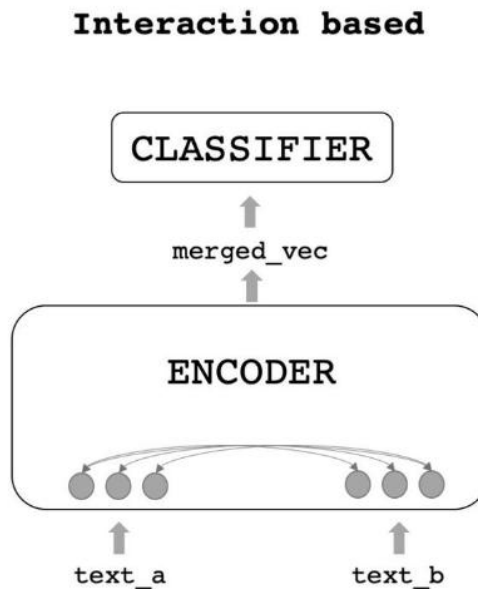


图 4. cross-encoder 架构

我们还将结果与词汇搜索（关键字搜索）进行比较。在这里，我们使用的是在 `rank bm25` 包中实现的 BM25 算法。

```

from rank_bm25 import BM25Okapi
from sklearn.feature_extraction import _stop_words
import string
from tqdm.autonotebook import tqdm
import numpy as np

# We lower case our text and remove stop-words from indexing
def bm25_tokenizer(text):
    tokenized_doc = []
    for token in text.lower().split():
        token = token.strip(string.punctuation)

        if len(token) > 0 and token not in _stop_words.ENGLISH_STOP_WORDS:
            tokenized_doc.append(token)
    return tokenized_doc

tokenized_corpus = []
for passage in tqdm(passages):
    tokenized_corpus.append(bm25_tokenizer(passage))

bm25 = BM25Okapi(tokenized_corpus)

```

图 4. BM25 算法代码实现

三、 实验流程

使用 retrieve_rerank_simple_wikipedia.ipynb 运行实验。

运行命令行 “!pip install sentence-transformers==2.2.2 rank_bm25==0.2.2” 下载 sentence-transformers 和 rank_bm25。

```

Collecting sentence-transformers==2.2.2Note: you may need to restart the kernel to use updated packages.
  Using cached sentence-transformers-2.2.2.tar.gz (85 kB)
  Preparing metadata (setup.py): started
  Preparing metadata (setup.py): finished with status 'done'
Collecting rank_bm25==0.2.2
  Using cached rank_bm25-0.2.2-py3-none-any.whl.metadata (3.2 kB)
Collecting transformers<5.0.0,>=4.6.0 (from sentence-transformers==2.2.2)
  Using cached transformers-4.41.0-py3-none-any.whl.metadata (43 kB)
Collecting tqdm (from sentence-transformers==2.2.2)
  Using cached tqdm-4.66.4-py3-none-any.whl.metadata (57 kB)
Collecting torch>=1.6.0 (from sentence-transformers==2.2.2)
  Using cached torch-2.3.0-cp311-cp311-win_amd64.whl.metadata (26 kB)
Collecting torchvision (from sentence-transformers==2.2.2)
  Using cached torchvision-0.18.0-cp311-cp311-win_amd64.whl.metadata (6.6 kB)
Requirement already satisfied: numpy in c:\users\yu939\appdata\local\packages\pythonsoftwarefoundation.pyth
Collecting scikit-learn (from sentence-transformers==2.2.2)
  Using cached scikit_learn-1.4.2-cp311-cp311-win_amd64.whl.metadata (11 kB)
Collecting scipy (from sentence-transformers==2.2.2)
  Using cached scipy-1.13.0-cp311-cp311-win_amd64.whl.metadata (60 kB)
Collecting nltk (from sentence-transformers==2.2.2)
  Using cached nltk-3.8.1-py3-none-any.whl.metadata (2.8 kB)
Collecting sentencepiece (from sentence-transformers==2.2.2)
  Using cached sentencepiece-0.2.0-cp311-cp311-win_amd64.whl.metadata (8.3 kB)
Collecting huggingface-hub>=0.4.0 (from sentence-transformers==2.2.2)
...

```

图 5. 配置环境

运行代码段下载数据集以及进行数据预处理。

```
import json
from sentence_transformers import SentenceTransformer, CrossEncoder, util
import grip
import os
import torch
import logging

print('cuda' if torch.cuda.is_available() else 'cpu')
# We use the Bi-Encoder to encode all passages, so that we can use it with semantic search
bi_encoder = SentenceTransformer('models/multi-qa-distilbert-cos-v1')
bi_encoder.max_seq_length = 256 #Truncate long passages to 256 tokens
top_k = 32 #Number of passages we want to retrieve with the bi-encoder

#The Bi-encoder will retrieve 100 documents. We use a cross-encoder, to re-rank the results list to improve the quality
cross_encoder = CrossEncoder('models/qnli-distilroberta-base')

# As dataset, we use Simple English Wikipedia. Compared to the full English wikipedia, it has only
# about 170k articles. We split these articles into paragraphs and encode them with the bi-encoder

wikipedia_filepath = 'data/simplewiki-2020-11-01.jsonl.gz'

if not os.path.exists(wikipedia_filepath):
    util.http_get('http://bert.net/datasets/simplewiki-2020-11-01.jsonl.gz', wikipedia_filepath)

passages = []
with grip.open(wikipedia_filepath, 'rt', encoding='utf8') as fin:
    for line in fin:
        data = json.loads(line.strip())
        #Add all paragraphs
        #passages.extend(data['paragraphs'])

        #Only add the first paragraph
        passages.append(data['paragraphs'][0])

print('Passages:', len(passages))

corpus_embeddings_path = 'data/corpus_embeddings.pt'
if os.path.exists(corpus_embeddings_path):
    corpus_embeddings = torch.load(corpus_embeddings_path)
else:
    logging.warning("Prebuilt corpus_embeddings not found. Please run the next block to compute your own corpus embeddings from scratch, which would take so much time to finish if you don't have a dedicated GPU.")
```

图 6. 数据预处理 1

```
# NOTE: only for those who want to compute your own corpus embeddings from scratch
# We encode all passages into our vector space.
corpus_embeddings = bi_encoder.encode(passages, convert_to_tensor=True, show_progress_bar=True)
```

图 7. 数据预处理 2

运行代码段实现 BM25 算法。

```
from rank_bm25 import BM25Okapi
from sklearn.feature_extraction import _stop_words
import string
from tqdm.autonotebook import tqdm
import numpy as np

# We lower case our text and remove stop-words from indexing
def bm25_tokenizer(text):
    tokenized_doc = []
    for token in text.lower().split():
        token = token.strip(string.punctuation)
        if len(token) > 0 and token not in _stop_words.ENGLISH_STOP_WORDS:
            tokenized_doc.append(token)
    return tokenized_doc

tokenized_corpus = []
for passage in tqdm(passages):
    tokenized_corpus.append(bm25_tokenizer(passage))

bm25 = BM25Okapi(tokenized_corpus)
```

100% | 169597/169597 [00:03<00:00, 53155.07it/s]

图 8. BM25 实现代码以及输出

运行代码段将搜索所有维基百科条目，以查找以下段落实现回答查询。


```
# This function will search all wikipedia articles for passages that
# answer the query
def search(query):
    print("Input question:", query)

    ##### BM25 search (lexical search) #####
    bm25_scores = bm25.get_scores(bm25_tokenizer(query))
    top_n = np.argsort(bm25_scores, -5)[-5:]
    bm25_hits = [{'corpus_id': idx, 'score': bm25_scores[idx]} for idx in top_n]
    bm25_hits = sorted(bm25_hits, key=lambda x: x['score'], reverse=True)

    print("Top-3 lexical search (BM25) hits")
    for hit in bm25_hits[0:3]:
        print("\t{:.3f}\t{}".format(hit['score'], passages[hit['corpus_id']].replace("\n", " ")))

    ##### Sematic Search #####
    # Encode the query using the bi-encoder and find potentially relevant passages
    question_embedding = bi_encoder.encode(query, convert_to_tensor=True)
    question_embedding = question_embedding.cuda()
    hits = util.semantic_search(question_embedding, corpus_embeddings, top_k=top_k)
    hits = hits[0] # Get the hits for the first query

    ##### Re-Ranking #####
    # Now, score all retrieved passages with the cross_encoder
    cross_inp = [[query, passages[hit['corpus_id']]] for hit in hits]
    cross_scores = cross_encoder.predict(cross_inp)

    # Sort results by the cross-encoder scores
    for idx in range(len(cross_scores)):
        hits[idx]['cross-score'] = cross_scores[idx]

    # Output of top-3 hits from bi-encoder
    print("\n-----\n")
    print("Top-3 Bi-Encoder Retrieval hits")
    hits = sorted(hits, key=lambda x: x['score'], reverse=True)
    for hit in hits[0:3]:
        print("\t{:.3f}\t{}".format(hit['score'], passages[hit['corpus_id']].replace("\n", " ")))

    # Output of top-3 hits from re-ranker
    print("\n-----\n")
    print("Top-3 Cross-Encoder Re-ranker hits")
    hits = sorted(hits, key=lambda x: x['cross-score'], reverse=True)
    for hit in hits[0:3]:
        print("\t{:.3f}\t{}".format(hit['cross-score'], passages[hit['corpus_id']].replace("\n", " ")))
```

图 9.search 函数的具体代码

四、 实验结果

通过询问不同问题可以得到相应的回答。

```
search(query = "What is the capital of the United States?")
Python

Input question: What is the capital of the United States?
Top-3 lexical search (BM25) hits
13.316 Capital punishment (the death penalty) has existed in the United States since before the United States was a country. As of 2017, capital punishment is legal in 30 of the 50 states. The federal government (in
11.434 Ohio is one of the 50 states in the United States. Its capital is Columbus. Columbus also is the largest city in Ohio.
11.179 Nevada is one of the United States' states. Its capital is Carson City. Other big cities are Las Vegas and Reno.

-----
Top-3 Bi-Encoder Retrieval hits
0.621 United States of America.
0.665 Cities in the United States:
0.553 Missouri is one of the 50 states in the United States. Its capital is Jefferson City. Its largest cities are Kansas City and Saint Louis. Some other cities are Columbia (which is where the University of Missou

-----
Top-3 Cross-Encoder Re-ranker hits
0.998 The continental United States is the area of the United States of America that is located in the continent of North America. It includes 49 of the 50 states (48 of which are located south of Canada and north o
0.998 Montana is a state in the United States. Its capital is Helena, and the largest city is Billings. Montana has many mountains in the western half of the state. Most of the eastern part of the state is prairie.
0.997 Nevada is one of the United States' states. Its capital is Carson City. Other big cities are Las Vegas and Reno.

search(query = "What is the best orchestra in the world?")
Python

Input question: What is the best orchestra in the world?
Top-3 lexical search (BM25) hits
15.328 The BBC Symphony Orchestra is the main orchestra of the British Broadcasting Corporation. It is one of the best orchestras in Britain.
15.330 The NHK Symphony Orchestra is a Japanese orchestra based in Tokyo, Japan. In Japanese it is written: NHK交響楽団, pronounced: Enueichikoi Kōkyō Gakudan. When the orchestra was started in 1926 it was called "N
14.079 The Bamberger Symphoniker (Bamberg Symphony Orchestra) is a world-famous orchestra from the city of Bamberg, Germany. It was formed in 1946. Most of the musicians who formed the orchestra were Germans who had

-----
Top-3 Bi-Encoder Retrieval hits
0.656 The BBC Symphony Orchestra is the main orchestra of the British Broadcasting Corporation. It is one of the best orchestras in Britain.
0.650 The Philharmonia is an orchestra from London.
0.624 The London Symphony Orchestra (LSO) is one of the most famous orchestras of the world. They are based in London's Barbican Centre, but they often tour to lots of different countries.

-----
Top-3 Cross-Encoder Re-ranker hits
0.918 The European Union Youth Orchestra (EUYO) is an orchestra for young people from countries in the European Union. The orchestra is given money by the European Union and by a number of EU member states. It give
0.908 The Berlin Philharmonic (in German: Die Berliner Philharmoniker), is an orchestra from Berlin, Germany. It is one of the greatest orchestras in the world. The conductor of the orchestra is Sir Simon Rattle.
0.897 The NHK Symphony Orchestra is a Japanese orchestra based in Tokyo, Japan. In Japanese it is written: NHK交響楽団, pronounced: Enueichikoi Kōkyō Gakudan. When the orchestra was started in 1926 it was called "N
```



```
search(query = "Number countries Europe")

Python

Input question: Number countries Europe
Top-3 lexical search (8025) hits
0.547 Ray Macdonald is a Scottish singer and songwriter. She became famous in 2007 with her first album "This Is The Life" and her first single "Poison Prince". She has become even more successful in Europe since h
13.758 The Croatian language is spoken mainly throughout the countries of Croatia and Bosnia and Herzegovina and in the surrounding countries of Europe.
13.019 Organization for Security and Co-operation in Europe (OSCE) is an international organization for peace and human rights. Presently, it has 57 countries as its members. Most of the member countries of the OSCE
-----

Top-3 BI-Encoder Retrieval hits
0.567 The Schengen Area is an area that includes 26 European countries. All of those countries have signed the Schengen Agreement in Schengen, Luxembourg in 1985.
0.545 As defined by Montevideo Convention, a country is a territory with distinct political boundaries that claim sovereignty over a specific geographic area with a permanent population, controlled by its own gove
0.526 The European Union (abbreviation: EU) is a confederation of 27 member countries in Europe established by the Maastricht Treaty in 1992-1993. The EU grew out of the European Economic Community (EEC) which was
-----

Top-3 Cross-Encoder Re-ranker hits
0.592 The Schengen Area is an area that includes 26 European countries. All of those countries have signed the Schengen Agreement in Schengen, Luxembourg in 1985.
0.585 Organization for Security and Co-operation in Europe (OSCE) is an international organization for peace and human rights. Presently, it has 57 countries as its members. Most of the member countries of the OSCE
0.577 The Concert of Europe was a group of countries in Europe who worked together and agreed on policies to maintain a steady balance of power which included alliances between the great powers in 1814 and 1914.
-----

search(query = "When did the cold war end?")

Python

Input question: When did the cold war end?
Top-3 lexical search (8025) hits
17.374 The Cold War was the tense relationship between the United States (and its allies), and the Soviet Union (the USSR and its allies) between the end of World War II and the fall of the Soviet Union. It is calle
17.291 The Reagan Doctrine was a document by the United States under the Reagan Administration. It was about being against the global influence of the Soviet Union during the final years of the Cold War. The doctrine
15.420 Cold Norton is a village and civil parish in Maldon District, Essex, England. In 2001 there were 1103 people living in Cold Norton. Cold Norton is at the south-east end of the Danbury Ridge.
-----

Top-3 BI-Encoder Retrieval hits
0.616 The Cold War was the tense relationship between the United States (and its allies), and the Soviet Union (the USSR and its allies) between the end of World War II and the fall of the Soviet Union. It is calle
0.543 A speech was made by American President Harry S. Truman to the U.S. Congress on 12 March 1947. In this speech he said he thought that the United States should help Greece and Turkey to stop them being 'Total
0.506 The Sino-Soviet split (1960-1989) was a time when the relations between the People's Republic of China and the Soviet Union weakened during the Cold War. Eventually, China's leader, Mao Zedong, decided to bre
-----

Top-3 Cross-Encoder Re-ranker hits
0.596 The Reagan Doctrine was a document by the United States under the Reagan Administration. It was about being against the global influence of the Soviet Union during the final years of the Cold War. The doctrine
0.594 The Cold War was the tense relationship between the United States (and its allies), and the Soviet Union (the USSR and its allies) between the end of World War II and the fall of the Soviet Union. It is calle
0.593 The 1950s was the decade that started on January 1, 1950, and ended on December 31, 1959.
-----

search(query = "How long do cats live?")

Python

Input question: How long do cats live?
Top-3 lexical search (8025) hits
22.997 Reliable information on the lifespans of house cats is hard to find. However, research has been done to get an estimate (an educated guess) on how long cats usually live. Cats usually live for 13 to 20 years.
16.574 The sabretoothed cats or sabretooth cats are some of the best known and most popular extinct animals. They are among the most impressive carnivores that ever have lived. These cats had long canines and jaws w
16.490 The Cyprus cat is a breed of cat. These cats are thought to have first come from ancient Egypt or Palestine. They were brought to the island of Cyprus by St. Helen. These are now common domestic cats that liv
-----

Top-3 BI-Encoder Retrieval hits
0.710 Reliable information on the lifespans of house cats is hard to find. However, research has been done to get an estimate (an educated guess) on how long cats usually live. Cats usually live for 13 to 20 years.
0.555 The Aethysthine python can live up to around the same amount of years as a ball python. The ball python can live up to 20-25 years depending on the habitat, weather and food source.
0.528 Bobcat ("lynx rufus") are fierce cats that live in forests, swamps, mountains, prairie, and deserts in much of North America. Bobcats are generally nocturnal (most active at night), but are most active at daw
-----

Top-3 Cross-Encoder Re-ranker hits
0.592 Pumas ("Puma concolor"), also called cougars, mountain lions or brown panthers, are large wild cats that live in the western half of North America, along with Florida, and most of Central and South America. P
0.591 Bobcat ("lynx rufus") are fierce cats that live in forests, swamps, mountains, prairie, and deserts in much of North America. Bobcats are generally nocturnal (most active at night), but are most active at daw
0.584 Reliable information on the lifespans of house cats is hard to find. However, research has been done to get an estimate (an educated guess) on how long cats usually live. Cats usually live for 13 to 20 years.
-----

search(query = "How many people live in Toronto?")

Python

Input question: How many people live in Toronto?
Top-3 lexical search (8025) hits
15.978 Markham, Ontario is a city in Regional Municipality of York, in the Greater Toronto Area of Southern Ontario, Canada. There are twice as many people there as in 1990. 261,573 people live in Markham. It is the
11.299 The Toronto Zoo is a zoo in Toronto, Ontario, Canada. With , the Toronto Zoo is the largest zoo in Canada.
10.679 Dennis Mui-man-wong ( 卞 文 彬 ) is a Canadian politician. He is a Toronto city councillor. He is the person that represents Ward 16, an area of Toronto. He is a chairperson of the Employee and Labour Relations C
-----

Top-3 BI-Encoder Retrieval hits
0.575 It has about 110,000 people living there.
0.500 Toronto is a city in Iowa in the United States.
0.494 Vaughan is a city in Ontario, Canada, 335,000 people live there .
-----

Top-3 Cross-Encoder Re-ranker hits
0.598 It has about 110,000 people living there.
0.508 It has about 100,000 inhabitants, a university and a well-known soccer team.
0.494 As of January 1, 2010, about 164,294 people lived there. That was 295.83 people per km². The total area is 555.35 km².
-----

search(query = "Oldest US president")

Python

Input question: Oldest US president
Top-3 lexical search (8025) hits
11.010 Glafcos Ioannou Clerides (; 24 April 1919 - 15 November 2013) was a Greek-Cypriot politician. He was the fourth President of Cyprus. He was the oldest living former President of the Republic of Cyprus.
9.217 José Celso de Mello Filho (Tufi, November 1, 1945). is a Brazilian jurist. He is the oldest member of the Supreme Federal Court of Brazil. He was nominated by President José Sarney in 1989.
8.172 USS "Constitution" is a wooden, three-masted heavy frigate of the United States Navy. Named by President George Washington after the Constitution of the United States of America, she is the world's oldest com
-----

Top-3 BI-Encoder Retrieval hits
0.591 Francisco Muñoz Olivera (13 December 1904 - 29 January 2018), age 113 years, 47 days, for a brief time was recognized as the oldest man in the world. He was born at Extremadura, and joined the army at 19 to fi
0.576 Gertrude Baines (April 6, 1894 - September 11, 2009) was an American supercentenarian. She was the oldest person in the world from January 2, 2009 until her death on September 11, 2009. She was the last perso
0.566 Jirosomen Kimura ( 木村 次郎右衛門, 19 April 1897 - 12 June 2013) was a Japanese supercentenarian recognized by the Guinness World Records as the oldest man ever as well as the only man who has become at least
-----

Top-3 Cross-Encoder Re-ranker hits
0.595 Richard Arden Overton (May 11, 1906 - December 27, 2018) was an American supercentenarian. He was the oldest verified surviving American World War II veteran, as well as the oldest American man. In 2013 he wa
0.589 Alben William Barkley (November 24, 1877 - April 30, 1956) was a Democratic member of the U.S. House of Representatives and the United States Senate from Paducah, Kentucky, majority leader of the Senate, and
0.588 Benjamin Harrison (August 20, 1833-March 13, 1901) was the 23rd president of the United States (1889-1893).
-----

search(query = "Coldest place earth")

Python

Input question: Coldest place earth
Top-3 lexical search (8025) hits
24.891 East Antarctica, also called Greater Antarctica, is the largest part (two-thirds) of the Antarctic continent. It is on the Indian Ocean side of the Transantarctic Mountains. It is the coldest, windiest, and d
12.650 Earth Day is a day that is supposed to inspire more awareness and appreciation for the Earth's natural environment. It takes place each year on April 22. It now takes place in more than 193 countries around th
12.172 Heinrich events occurred during the coldest point of "Bond Cycles" in which many icebergs were discharged into the North Atlantic and melted.
-----

Top-3 BI-Encoder Retrieval hits
0.624 East Antarctica, also called Greater Antarctica, is the largest part (two-thirds) of the Antarctic continent. It is on the Indian Ocean side of the Transantarctic Mountains. It is the coldest, windiest, and d
0.526 Places with a subarctic climate (also called boreal climate) have long, usually very cold winters, and short, warm summers. It is found on large landmasses, away from oceans, usually at latitudes from 50° to
0.522 The North Pole is the point that is farthest north on Earth. It is the point on which axis of Earth turns. It is in the Arctic Ocean and it is cold there because the sun does not shine there for about half a
-----

Top-3 Cross-Encoder Re-ranker hits
0.589 East Antarctica, also called Greater Antarctica, is the largest part (two-thirds) of the Antarctic continent. It is on the Indian Ocean side of the Transantarctic Mountains. It is the coldest, windiest, and d
0.584 Summer is one of the four seasons. It is the hottest season of the year. In some places, summer is the wettest season (with the most rain), and in 'other places', it is a dry season. Four seasons are found in
0.568 The stratospaue is the part of the atmosphere between the stratosphere and mesosphere. In the stratosphere the temperature gets higher the higher you go. The stratospaue is where it is warmest. On Earth,
-----
```

```
search(query = "Elon Musk year birth")
Python

Input question: Elon Musk year birth
Top-3 lexical search (BM25) hits
23.364 Tesla, Inc. is a company based in Palo Alto, California which makes electric cars. It was started in 2003 by Martin Eberhard, Dylan Stott, and Elon Musk (who also co-founded PayPal and SpaceX and is the CEO o
19.945 The Boring Company is a tunnel boring company founded by Elon Musk, who earlier started SpaceX. It aims to reduce traffic congestion in urban areas. It is involved in the building of the Hyperloop in Los Ange
18.392 Elon Reeve Musk (born June 28, 1971) is a businessman and philanthropist. He was born in South Africa. He moved to Canada and later became an American citizen. Musk is the current CEO & Chief Product Architect

-----
Top-3 Bi-Encoder Retrieval hits
0.494 Donald Lusk (October 28, 1913 - December 30, 2018) was an American retired animator and director. He worked for The Walt Disney Company.
0.487 Elon Reeve Musk (born June 28, 1971) is a businessman and philanthropist. He was born in South Africa. He moved to Canada and later became an American citizen. Musk is the current CEO & Chief Product Architect
0.441 Uilly Murs (born 14 May 1984) is an English singer. He comes from Witham, Essex. He became famous after he finished second in The X Factor in 2009.

-----
Top-3 Cross-Encoder Re-ranker hits
0.997 Kim Hye-chul (born July 10, 1983)
0.996 Yang Hyong-sop (born 1 October 1925)
0.996 He was born in 1680 in the city of Uch, during the Mughal Empire (present day Pakistan). After his early education, he went to Lahore where he met Inayat Arian, and became his student.

search(query = "Paris eiffel tower")
Python

Input question: Paris eiffel tower
Top-3 lexical search (BM25) hits
27.300 The Eiffel Tower (French: La Tour Eiffel, , IPA pronunciation: "EYF-ful" English; "eh-FEHL" French) is a landmark in Paris. It was built between 1887 and 1889 for the Exposition Universelle (World Fair). The
25.263 Paris is a city in the U.S. state of Texas. It is in Lamar County, Texas. It had a population of 25,171 in 2010. It has been called the "Second Largest Paris in the World". It has a replica of the Eiffel Tower
24.659 Paris is a city in the U.S. state of Tennessee. It had a population of 25,171 in 2010. It has been called the "World's Biggest Fish Fry". It has a 70-foot replica of the Eiffel Tower.

-----
Top-3 Bi-Encoder Retrieval hits
0.780 The Eiffel Tower (French: La Tour Eiffel, , IPA pronunciation: "EYF-ful" English; "eh-FEHL" French) is a landmark in Paris. It was built between 1887 and 1889 for the Exposition Universelle (World Fair). The
0.554 Alexandre Gustave Eiffel (December 15, 1832 - December 27, 1923; , ) was a French structural engineer and architect. He is known for designing the Eiffel Tower. He also designed the armature (supporting frame
0.512 Paris Square (, "Kikar Pariz", also called: France Square - "Kikar Tzarfat") is a square or plaza in Jerusalem. The name "France Square" was established in 1959. In 2007 the city of Paris gave Jerusalem a "fr

-----
Top-3 Cross-Encoder Re-ranker hits
0.972 The Eiffel Tower (French: La Tour Eiffel, , IPA pronunciation: "EYF-ful" English; "eh-FEHL" French) is a landmark in Paris. It was built between 1887 and 1889 for the Exposition Universelle (World Fair). The
0.969 Paris is a city in the U.S. state of Tennessee. It had a population of 25,171 in 2010. It has been called the "World's Biggest Fish Fry". It has a 70-foot replica of the Eiffel Tower.
0.789 Alexandre Gustave Eiffel (December 15, 1832 - December 27, 1923; , ) was a French structural engineer and architect. He is known for designing the Eiffel Tower. He also designed the armature (supporting frame

search(query = "Which US president was killed?")
Python

Input question: Which US president was killed?
Top-3 lexical search (BM25) hits
18.179 Lyndon Baines Johnson (August 27, 1908 - January 22, 1973) was a member of the Democratic Party and the 36th president of the United States serving from 1963 to 1969. Johnson took over as president when Presi
18.091 Lech Kaczynski, the fourth President of the Republic of Poland, died on 10 April 2010. He died in a plane crash outside of Smolensk, Russia. The plane was a Tu-154 belonging to the Polish Air Force. The crash
9.791 Jacobo Majluta Azar (October 9, 1934 - March 2, 1996) was a Dominican politician. He was Vice President of the Dominican Republic during the Antonio Guzmán Fernández presidency between 1978 to 1982. He became

-----
Top-3 Bi-Encoder Retrieval hits
0.652 William McKinley, the 25th President of the United States, was assassinated on September 6, 1901, inside the Temple of Music on the grounds of the Pan-American Exposition in Buffalo, New York.
0.621 John F. Kennedy was the 35th President of the United States. He was assassinated (murdered) in Dealey Plaza, Dallas, Texas, on Friday, November 22, 1963. This happened while he was traveling in a Presidential
0.615 The assassination of Abraham Lincoln was one of the last events in the American Civil War, and happened on Good Friday, April 14, 1865. President Lincoln was shot while watching the play "Our American Cousin"

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Top-3 Cross-Encoder Re-ranker hits
0.999 John Wilkes Booth (May 10, 1838 - April 26, 1865) was an American actor who assassinated U.S. president Abraham Lincoln on April 14, 1865, in Washington, D.C. Lincoln died the next morning. Booth was born in
0.998 Lee Harvey Oswald (October 18, 1939 - November 24, 1963) was, according to five government investigations, the sniper who assassinated U.S. President John F. Kennedy in Dallas, Texas on November 22, 1963.
0.998 The assassination of Robert F. Kennedy, a United States Senator and brother of assassinated President John F. Kennedy, took place shortly after midnight on June 5, 1968, in Los Angeles after winning his camp

search(query="When is Chinese New Year")
Python

Input question: When is Chinese New Year
Top-3 lexical search (BM25) hits
18.743 Chinese New Year, known in China as the Springfestival and in Singapore as the LunarNewYear, is a holiday on and around the new moon on the first day of the year in the traditional Chinese calendar. This calen
18.527 New Year in Japan is one of the most important festivals. Unlike the Chinese New Year, it is held on January 1.
15.789 The CCTV New Year's Gala (Simplified Chinese: 中国中央电视台春节联欢晚会; Traditional Chinese: 中國中央電視台春節聯歡晚會; Pinyin: "Zhōngguó zhōngyǎng diànshìtái chūnjié liánhuān wǎnhuì") is a Chinese New Year

-----
Top-3 Bi-Encoder Retrieval hits
0.725 Chinese New Year, known in China as the Springfestival and in Singapore as the LunarNewYear, is a holiday on and around the new moon on the first day of the year in the traditional Chinese calendar. This calen
0.637 Chinese National Day is the national day of China. It may mean:
0.598 National Day is a yearly holiday in the People's Republic of China. It celebrates the beginning of its new government on October1, 1949. It is one of two Golden Weeks in the country, along with the Chinese Ne

-----
Top-3 Cross-Encoder Re-ranker hits
0.942 New Year in Japan is one of the most important festivals. Unlike the Chinese New Year, it is held on January 1.
0.931 Chinese New Year, known in China as the Springfestival and in Singapore as the LunarNewYear, is a holiday on and around the new moon on the first day of the year in the traditional Chinese calendar. This calen
0.835 National Day is a yearly holiday in the People's Republic of China. It celebrates the beginning of its new government on October1, 1949. It is one of two Golden Weeks in the country, along with the Chinese Ne
```

图 10. 查询问题及结果

五、 结果分析

这段代码实现了一个用于在维基百科文章中搜索回答特定问题的函数。输入问题后，首先使用 BM25 算法对问题进行词汇搜索，得到与问题相关的文章的分數。选择前 5 个分數最高的文章，将选择的文章的索引和分數存储在字典中，按照分數降序排序选定的文章，打印词汇搜索的前三个最相关文章的标题，遍历前三个最相关的文章，打印文章的分數和内容。以“query = "What is the capital of the United States?"”为例。

```

Top-3 lexical search (BM25) hits
13.316 Capital punishment (the death penalty) has existed in the United States since b
11.434 Ohio is one of the 50 states in the United States. Its capital is Columbus. Col
11.179 Nevada is one of the United States' states. Its capital is Carson City. Other b

```

图 11. 词汇搜索的前三个最相关文章的分数及内容

从图 11 可以看到,我们获得的前三个最相关的文章分数分别为 13.316、11.434、11.179。

然后使用双编码器对问题进行编码,以便进行语义搜索。使用语义搜索找到与问题语义相关的文章,获得使用双编码器得到的前三个最相关文章的分数及内容。

```

Top-3 Bi-Encoder Retrieval hits
0.621 United States of America;
0.605 Cities in the United States:
0.553 Missouri is one of the 50 states in the United States. Its capital is Jefferson City. Its largest cit

```

图 12. 使用双编码器得到的前三个最相关文章的分数及内容

使用交叉编码器对输入进行预测,得到前三个最相关文章的分数以及内容。

```

Top-3 Cross-Encoder Re-ranker hits
0.998 The continental United States is the area of the United States of America that is located in the continent of North America.
0.998 Montana is a state in the United States. Its capital is Helena, and the largest city is Billings. Montana has many mountains
0.997 Nevada is one of the United States' states. Its capital is Carson City. Other big cities are Las Vegas and Reno.

```

图 13. 使用交叉编码器得到的前三个最相关文章的分数及内容

通过实验结果可以看出实现了一个多阶段的搜索过程,首先进行词汇搜索,然后进行语义搜索,并最终通过交叉编码器进行结果的重新排序。

六、 心得体会

通过本次实验,我对信息检索系统有了一定的了解,通过研读代码以及对应的论文,对模型的架构有了清晰的认识。

在这个实验中,我实现了一个信息检索模型。在代码实现过程中,首先我需要对数据集进行预处理,需要安装模型的架构选择合适的数据集。在这个实验中,使用的数据集为简单维基百科[Simple Wikipedia],这是我第一次接触这个数据集,通过查阅资料对该数据集有了一定的了解。在这里之所以选择简单的,规模小的数据集是由于我们使用的语义相似度计算模型是 cross-encoder,适合用于少量候选问句场景。通过搜索资料与理解,对 multi-qa-distilbert-cos-v1 模型、cross-encoder 以及 BM25 算法有了一定的认识与了解,掌握了如何将其应用于信息检索任务中。

在进行代码运行的过程中,我们使用了 Jupyter Notebook 进行一步一步命令

行的运行。**Jupyter Notebook** 是一个非常强大的工具，常用于交互式地开发和展示数据科学项目。它将代码和它的输出集成到一个文档中，并且结合了可视的叙述性文本、数学方程和其他丰富的媒体。它直观的工作流促进了迭代和快速的开发，使得 **notebook** 在当代数据科学、分析和越来越多的科学研究中越来越受欢迎。最重要的是，作为开源项目的一部分，它们是完全免费的。

通过本次实验激发了我对信息检索系统的兴趣，通过研读代码，学习新的模型架构，提高了阅读和理解代码的能力。同时，将理论知识落实到实践，也大大积累了我的实践经验，学会了使用 **Jupyter Notebook** 等工具来简化实验流程和提高效率，为今后的学习奠定基础，提高学习效率。